Polygons & Quadrilaterals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Proving triangles similar

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

2 [Foundational] • Target: Proving triangles similar

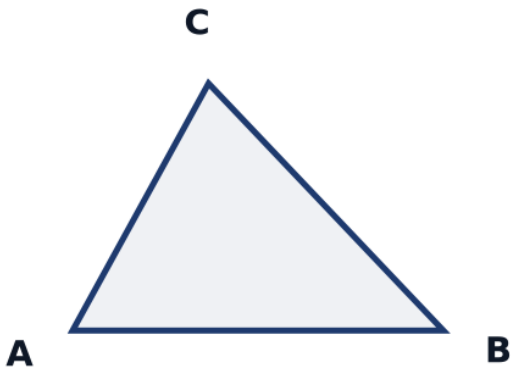
On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

3 [Foundational] • Target: Proving triangles similar

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

4 [Foundational] • Target: Proving triangles similar

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.



5 [Foundational] • Target: Proving triangles similar

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

6 [Foundational] • Target: Proving triangles similar

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

7 [Foundational] • Target: Proving triangles similar

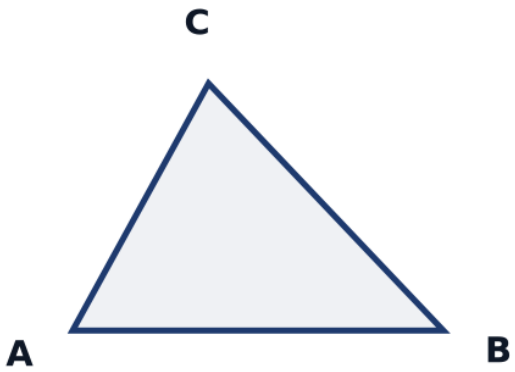
A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

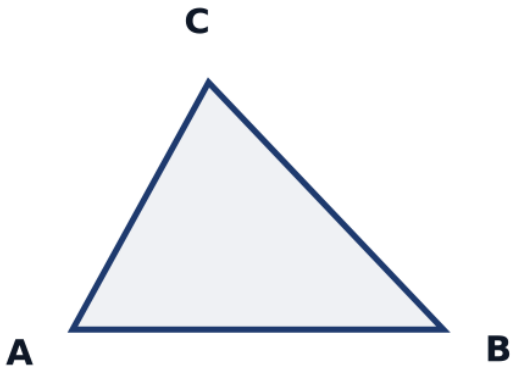
8 **[Intermediate]** • **Target: Proving triangles similar**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



9 **[Intermediate]** • **Target: Proving triangles similar**

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



10 **[Intermediate]** • **Target: Proving triangles similar**

A geometric figure scaled by factor $k = 5$ has an original side length of $7\ \text{cm}$. What is the corresponding image length? Show your complete algebraic work.

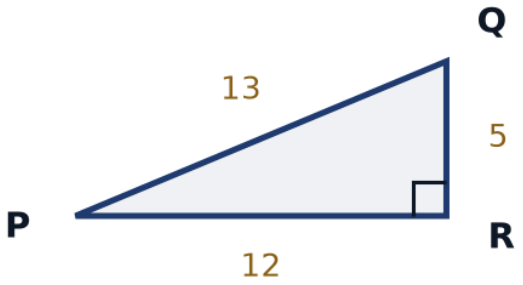
11 **[Intermediate]** • **Target: Proving triangles similar**

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

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12 [Intermediate] • Target: Applications of similar triangles

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



13 [Intermediate] • Target: Applications of similar triangles

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

14 [Intermediate] • Target: Applications of similar triangles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

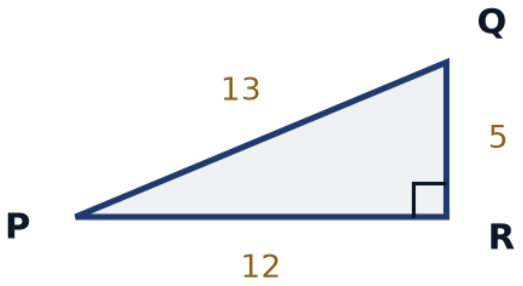
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 10.01 Coordinate Geometry

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

16 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.



17 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

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18 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 3.01 Triangles & Congruence

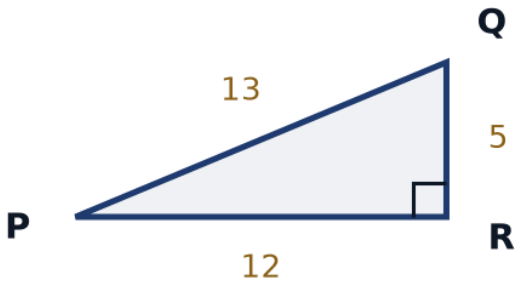
In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

19 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

20 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 5.01 Right Triangles & Trigonometry

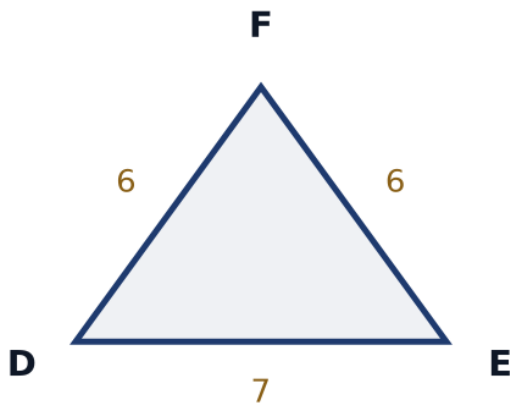
On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Review: Similar figures

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 99°

b. 109°

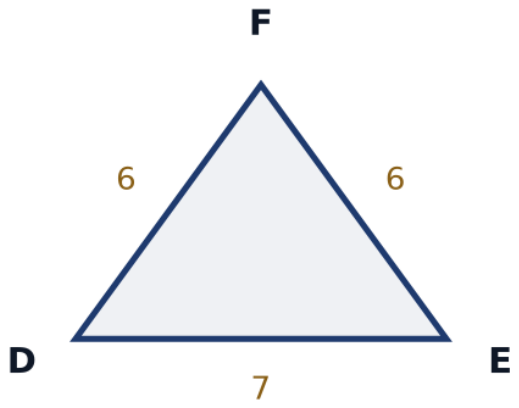
c. 91°

d. 144°

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22 [Foundational] • Target: Review: Similar figures

A geometric figure scaled by factor $k = 3$ has an original side length of 5 cm. What is the corresponding image length?



a. 12 cm

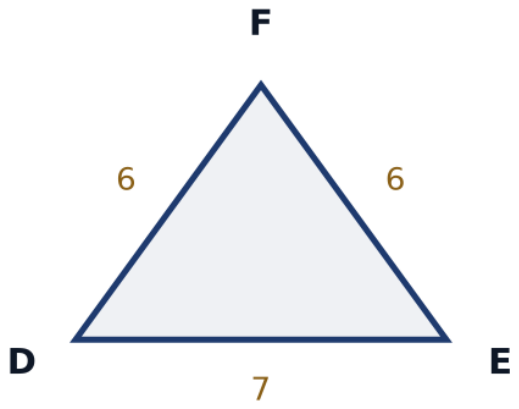
b. 15 cm

c. 20 cm

d. 45 cm

23 [Foundational] • Target: Review: Similar figures

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?



a. 16

b. 23

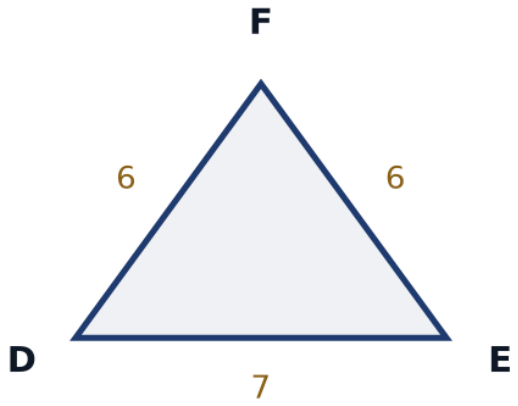
c. 21

d. 26

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24 [Foundational] • Target: Review: Similar figures

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 78°

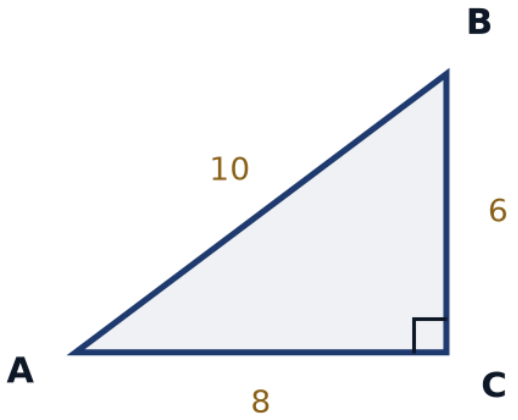
b. 88°

c. 70°

d. 144°

25 [Foundational] • Target: Review: Identifying similar triangles

In ΔABC studying review: identifying similar triangles, $m\angle A = 37^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 98°

b. 108°

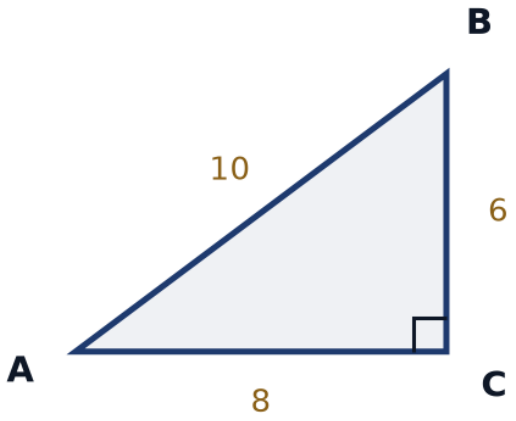
c. 90°

d. 143°

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26 [Foundational] • Target: Review: Identifying similar triangles

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

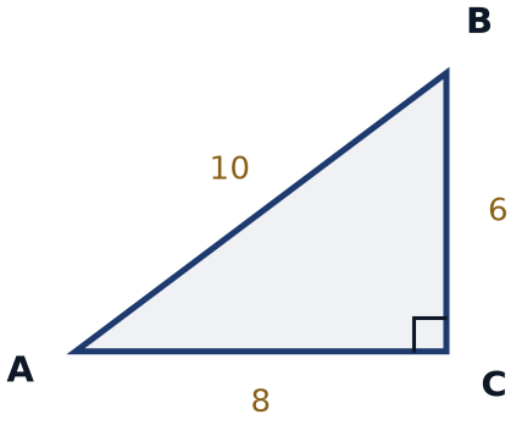
b. 18 cm

c. 24 cm

d. 54 cm

27 [Foundational] • Target: Review: Identifying similar triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?



a. 11

b. 15

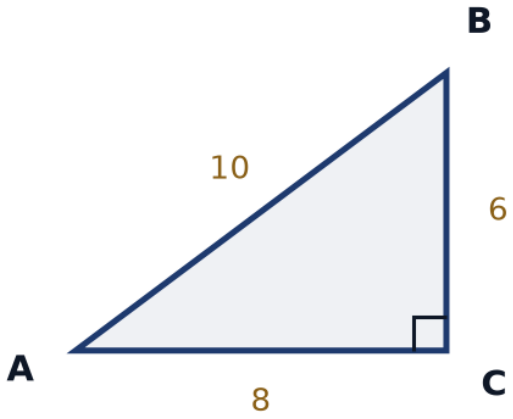
c. 13

d. 15

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28 [Foundational] • Target: Review: Identifying similar triangles

In ΔABC studying review: identifying similar triangles, $m\angle A = 37^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 77°

b. 87°

c. 69°

d. 143°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Review: Similar figures

A geometric figure scaled by factor $k = 2$ has an original side length of 5 cm. What is the corresponding image length?

a. 8 cm

b. 10 cm

c. 15 cm

d. 20 cm

30 [Intermediate] • Target: Review: Similar figures

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 92°

b. 102°

c. 84°

d. 144°

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31 [Intermediate] • Target: Review: Similar figures

In ΔABC studying review: similar figures, $m\angle A = 36^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 85°

b. 95°

c. 77°

d. 144°

32 [Intermediate] • Target: Review: Similar figures

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 6)$. What is its y-coordinate when $x = 4$?

a. 21

b. 28

c. 26

d. 31

33 [Intermediate] • Target: Review: Identifying similar triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 1$?

a. 7

b. 11

c. 9

d. 11

34 [Intermediate] • Target: Review: Identifying similar triangles

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 7)$. What is its y-coordinate when $x = 2$?

a. 9

b. 13

c. 11

d. 13

35 [Intermediate] • Target: Review: Identifying similar triangles

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

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36 [Intermediate] • Target: Proving triangles similar

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

37 [Intermediate] • Target: Proving triangles similar

In ΔABC studying proving triangles similar, $m\angle A = 38^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 90°

b. 100°

c. 82°

d. 142°

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Proving triangles similar • Cross-Lesson: 3.01 Triangles & Congruence

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

39 [Challenging] • Target: Proving triangles similar • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 17

b. 22

c. 20

d. 23

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40 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 5.01 Right Triangles & Trigonometry

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 15

c. 13

d. 17

41 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 7.01 Circles

In ΔABC studying applications of similar triangles, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 82°

b. 92°

c. 74°

d. 141°

42 [Challenging] • Target: Applications of similar triangles • Cross-Lesson: 8.01 Area, Surface Area & Volume

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

c. 48 cm

d. 200 cm

43 [Challenging] • Target: Perimeters and areas of similar figures • Cross-Lesson: 9.01 Transformations & Symmetry

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length?

a. 6 cm

b. 8 cm

c. 12 cm

d. 16 cm

44 [Challenging] • Target: Perimeters and areas of similar figures • Cross-Lesson: 1.01 Foundations & Reasoning

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 3)$. What is its y-coordinate when $x = 2$?

a. 8

b. 15

c. 13

d. 18

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45 [Challenging] • Target: Perimeters and areas of similar figures • Cross-Lesson: 10.01 Coordinate Geometry

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length?

a. 12 cm

b. 16 cm

c. 20 cm

d. 64 cm

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